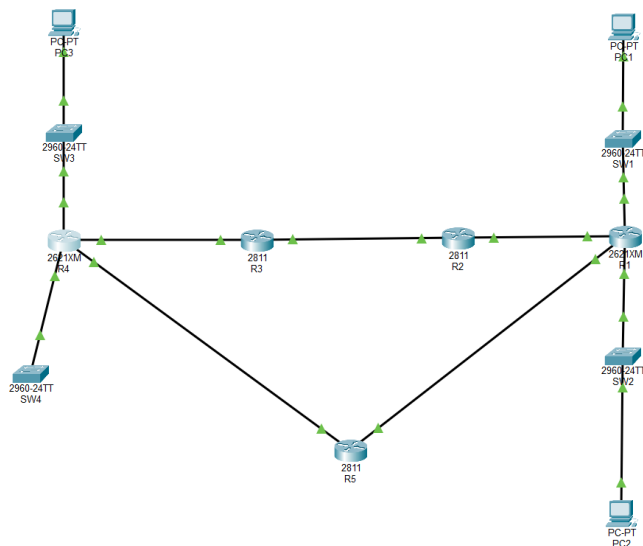


Practical Week 8

Topology



Task 74

```
R1(config)# router rip
R1(config-router)# network 10.0.1.1
R1(config-router)#network 10.0.0.1
R1(config-router)#network 10.0.2.1
R1(config-router)#network 10.0.3.1
R2(config-router)# network 10.0.0.2
R2(config-router)# network 10.1.0.2
R2(config-router)# no auto summary
R4(config)# router rip
R4(config-router)#network 10.1.1.1
R4(config-router)#network 10.1.2.1
R4(config-router)#network 10.1.3.1
R3(config-router)#network 10.1.0.1
R3(config-router)#network 10.1.1.2
R4(config-router)#network 203.0.113.1
R4(config-router)# no auto-summary

R5(config)# router rip
R5(config-router)#network 10.1.3.2
R5(config-router)#network 10.0.3.2
R5(config-router)#no auto-summary
```

Task 75

```

RIP: sending v1 update to 255.255.255.255 via FastEthernet1/0 (10.0.2.1)
RIP: build update entries
    network 10.0.0.0 metric 1
    network 10.0.1.0 metric 1
    network 10.0.3.0 metric 1
    network 10.1.0.0 metric 2
    network 10.1.1.0 metric 3
    network 10.1.2.0 metric 3
    network 10.1.3.0 metric 2
    network 203.0.113.0 metric 3
RIP: sending v1 update to 255.255.255.255 via FastEthernet1/1 (10.0.3.1)
RIP: build update entries
    network 10.0.0.0 metric 1
    network 10.0.1.0 metric 1
    network 10.0.2.0 metric 1
    network 10.1.0.0 metric 2
RIP: received v1 update from 10.0.3.2 on FastEthernet1/1
    10.1.1.0 in 2 hops
    10.1.2.0 in 2 hops
    10.1.3.0 in 1 hops
    203.0.113.0 in 2 hops
RIP: received v1 update from 10.0.0.2 on FastEthernet0/0
    10.1.0.0 in 1 hops
    10.1.1.0 in 2 hops
    10.1.2.0 in 3 hops
    203.0.113.0 in 3 hops
RIP: sending v1 update to 255.255.255.255 via FastEthernet0/0 (10.0.0.1)
RIP: build update entries
    network 10.0.1.0 metric 1
    network 10.0.2.0 metric 1
    network 10.0.3.0 metric 1
    network 10.1.2.0 metric 3
    network 10.1.3.0 metric 2
    network 203.0.113.0 metric 3
RIP: sending v1 update to 255.255.255.255 via FastEthernet0/1 (10.0.1.1)
RIP: build update entries
    network 10.0.0.0 metric 1
    network 10.0.2.0 metric 1
    network 10.0.3.0 metric 1
    network 10.1.0.0 metric 2
    network 10.1.1.0 metric 3
    network 10.1.2.0 metric 3
    network 10.1.3.0 metric 2
    network 203.0.113.0 metric 3
RIP: sending v1 update to 255.255.255.255 via FastEthernet1/0 (10.0.2.1)
RIP: build update entries
    network 10.0.0.0 metric 1
    network 10.0.1.0 metric 1
    network 10.0.3.0 metric 1
    network 10.1.0.0 metric 2
    network 10.1.1.0 metric 3
    network 10.1.2.0 metric 3
    network 10.1.3.0 metric 2
    network 203.0.113.0 metric 3
RIP: sending v1 update to 255.255.255.255 via FastEthernet1/1 (10.0.3.1)

```

Task 76

```

R1(config-router)# version 2
R2(config)# router rip
R3(config)#router rip
R2(config-router)# version 2
R3(config-router)# version 2
R4(config)# router rip
R5(config)# router rip
R4(config-router)# version 2
R5(config-router)# version 2

```

Task 77

```
R1# u all
```

Task 78

```

-----
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

```

Gateway of last resort is not set

```

      10.0.0.0/24 is subnetted, 8 subnets
C       10.0.0.0 is directly connected, FastEthernet0/0
C       10.0.1.0 is directly connected, FastEthernet0/1
C       10.0.2.0 is directly connected, FastEthernet1/0
C       10.0.3.0 is directly connected, FastEthernet1/1
R       10.1.0.0 [120/1] via 10.0.0.2, 00:00:21, FastEthernet0/0
R       10.1.1.0 [120/2] via 10.0.0.2, 00:00:21, FastEthernet0/0
          [120/2] via 10.0.3.2, 00:00:10, FastEthernet1/1
R       10.1.2.0 [120/2] via 10.0.3.2, 00:00:10, FastEthernet1/1
R       10.1.3.0 [120/1] via 10.0.3.2, 00:00:10, FastEthernet1/1
R       203.0.113.0/24 [120/2] via 10.0.3.2, 00:00:10, FastEthernet1/1

```

Task 79

```

R1# show ip rip database
10.0.0.0/24      auto-summary
10.0.0.0/24      directly connected, FastEthernet0/0
10.0.1.0/24      auto-summary
10.0.1.0/24      directly connected, FastEthernet0/1
10.0.2.0/24      auto-summary
10.0.2.0/24      directly connected, FastEthernet1/0
10.0.3.0/24      auto-summary
10.0.3.0/24      directly connected, FastEthernet1/1
10.1.0.0/24      auto-summary
10.1.0.0/24      [1] via 10.0.0.2, 00:00:16, FastEthernet0/0
10.1.1.0/24      auto-summary
10.1.1.0/24      [2] via 10.0.0.2, 00:00:16, FastEthernet0/0      [2] via 10.0.3.2, 00:00:29, FastEthernet1/1
10.1.2.0/24      auto-summary
10.1.2.0/24      [2] via 10.0.3.2, 00:00:29, FastEthernet1/1
10.1.3.0/24      auto-summary
10.1.3.0/24      [1] via 10.0.3.2, 00:00:29, FastEthernet1/1
203.0.113.0/24   auto-summary
203.0.113.0/24   [2] via 10.0.3.2, 00:00:29, FastEthernet1/1

```

Task 80

```

R1(config)# router ospf 1
R1(config-router)#network 10.0.1.1 0.255.255.255 area 0
R1(config-router)#network 10.0.0.1 0.255.255.255 area 1
R1(config-router)#network 10.0.2.1 0.255.255.255 area 0
R1(config-router)#network 10.0.3.1 0.255.255.255 area 0
R2(config-router)#network 10.0.0.2 0.255.255.255 area 0
R2(config-router)#
01:57:23: %OSPF-5-ADJCHG: Process 1, Nbr 10.0.3.1 on Fa.

R2(config-router)#network 10.1.0.2 0.255.255.255 area 0

R3(config)# router ospf 1
R3(config-router)#network 10.1.0.1 0.255.255.255 area 0
R3(config-router)#network
01:59:48: %OSPF-5-ADJCHG: Process 1, Nbr 10.1.0.2 on Fa.

% Incomplete command.
R3(config-router)#network 10.1.1.2 0.255.255.255 area 0

R4(config-router)#network 10.1.1.1 0.255.255.255 area 0

R4(config-router)#network 10.1.2.1 0.255.255.255 area 0
R4(config-router)#network 10.1.3.1 0.255.255.255 area 0
R4(config-router)#network 203.0.113.2 0.255.255.255 area 0

```

```

R5(config)# router ospf 1
R5(config-router)#network 10.1.3.2 0.255.255.255 area 0
R5(config-router)#
02:08:49: %OSPF-5-ADJCHG: Process 1, Nbr 10.0.3.1 on Fa:
network 10.1.
02:08:56: %OSPF-5-ADJCHG: Process 1, Nbr 203.0.113.1 on
3.2 0.255.255.255 area 0

```

Task 81

```

R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

```

Gateway of last resort is not set

```

      10.0.0.0/24 is subnetted, 8 subnets
C       10.0.0.0 is directly connected, FastEthernet0/0
C       10.0.1.0 is directly connected, FastEthernet0/1
C       10.0.2.0 is directly connected, FastEthernet1/0
C       10.0.3.0 is directly connected, FastEthernet1/1
O       10.1.0.0 [110/2] via 10.0.0.2, 00:16:17, FastEthernet0/0
O       10.1.1.0 [110/3] via 10.0.0.2, 00:03:06, FastEthernet0/0
        [110/3] via 10.0.3.2, 00:03:06, FastEthernet1/1
O       10.1.2.0 [110/3] via 10.0.3.2, 00:03:06, FastEthernet1/1
O       10.1.3.0 [110/2] via 10.0.3.2, 00:03:06, FastEthernet1/1
O       203.0.113.0/24 [110/3] via 10.0.3.2, 00:03:06, FastEthernet1/1

```

Task 82

FastEthernet0/0	
Port Status	☐ On

```

R1# show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

```

Gateway of last resort is not set

```

      10.0.0.0/24 is subnetted, 7 subnets
C       10.0.1.0 is directly connected, FastEthernet0/1
C       10.0.2.0 is directly connected, FastEthernet1/0
C       10.0.3.0 is directly connected, FastEthernet1/1
O       10.1.0.0 [110/4] via 10.0.3.2, 00:00:54, FastEthernet1/1
O       10.1.1.0 [110/3] via 10.0.3.2, 00:00:54, FastEthernet1/1
O       10.1.2.0 [110/3] via 10.0.3.2, 00:05:46, FastEthernet1/1
O       10.1.3.0 [110/2] via 10.0.3.2, 00:05:46, FastEthernet1/1
O       203.0.113.0/24 [110/3] via 10.0.3.2, 00:05:46, FastEthernet1/1

```

Task 83

```

R1# show ip ospf database
      OSPF Router with ID (10.0.3.1) (Process ID 1)

      Router Link States (Area 0)

Link ID        ADV Router    Age          Seq#          Checksum Link count
10.1.1.2       10.1.1.2       820          0x80000004   0x005c94 2
10.1.3.2       10.1.3.2       439          0x80000004   0x00bc29 2
203.0.113.1    203.0.113.1    439          0x80000006   0x008697 4
10.0.3.1       10.0.3.1       152          0x80000008   0x00da01 3
10.1.0.2       10.1.0.2       152          0x80000005   0x00cb51 1

      Net Link States (Area 0)

Link ID        ADV Router    Age          Seq#          Checksum
10.1.0.2       10.1.0.2     1054         0x80000001   0x00af16
10.1.1.2       10.1.1.2     820          0x80000001   0x007cd5
10.0.3.1       10.0.3.1     447          0x80000002   0x008d35
10.1.3.1       203.0.113.1  439          0x80000001   0x00855b

      Router Link States (Area 1)

Link ID        ADV Router    Age          Seq#          Checksum Link count
10.0.3.1       10.0.3.1     152          0x80000005   0x00f0fb 3

```

Task 84

It uses multicast traffic.

Task 85

So that they could be connected to the OSPF at the same time.

Task 86

They are still present because they are permanently there.

Task 87

The other address from the 10.1.1.0 route is absent

Task 88

RIP database does not support VLSM while OSPF supports VLSM